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SUPERSEDING
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MILITARY SPECIFICATION

CRANE, CRAWLER MOUNTED, 40 TON AT 12 FOOT RADIUS, 2 CUBIC YARD, DIESEL-ENGINE-DRIVEN

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 SCOPE. This specification covers a diesel-engine-driven, crawler-mounted crane suitable for use as a lifting crane with the capability of being converted for clamshell, dragline, shovel, backhoe, wrecking ball, or piledriver operation. The cranes provided shall have a rated lifting capacity of 40 tons at a 12 foot radius.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-W-410	- Wire Rope and Strand.
VV-F-800	- Fuel Oil, Diesel.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-V-173	- Varnish, Moisture and Fungus Resistant (For Treatment of Communications, Electronic and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-L-2104	- Lubricating Oil, Internal-Combustion Engine, Tactical Service.
MIL-L-2105	- Lubricating Oil, Gear, Multipurpose.
MIL-C-3580	- Cranes and Crane-Shovels, Truck, Crawler, and Wheel Mounted, Full-Revolving, and Their Attachments, Packaging of.
MIL-G-3859	- Grease Gun, Hand-Operated, Lever, Push and Screw Type.
MIL-E-10062	- Engine, Preparation for Shipment and Storage of.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-L-11025	- Lead, Pile-Driver; with Catwalk.
MIL-B-16028	- Bucket, Clamshell.
MIL-B-16029	- Bucket, Dragline.
MIL-H-21103	- Hammers, Piledriver, Self-Powered, Diesel.
MIL-B-23276	- Balls, Wrecking.
MIL-A-46153	- Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.
MIL-B-46176	- Brake Fluid, Silicone, Automotive, All-Weather, Operational and Preservative.
MIL-T-52176	- Teeth, Surface Ripping; (Points, Shanks, Keepers and Keeper-Locks) Bucket and Dipper, Crane Shovel.
MIL-T-52198	- Taglines, Crane and Crane-Shovel, Spring Loaded Reel Type.

STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-642	- Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1474	- Noise Limits for Army Materiel.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity and as directed by the contracting officer.)

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

DEPARTMENT OF LABOR (DOL)

OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Occupational Safety and Health Standards with Supplemental Revisions.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

ENVIRONMENTAL PROTECTION AGENCY

Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

TECHNICAL INFORMATION PACKAGE (TIP)

(Application for copies should be addresed to US Army Mobility Equipment Research and Development Command, ATTN: DRDME-HK, Fort Belvoir, VA 22060.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC, (ANSI).

B30.5 - Safety Code for Crawler, Locomotive, and Truck Cranes.

(Application for copies should be addressed to the American National Standards Institute Inc., 1430 Broadway, New York, NY 10018.)

ASSOCIATION OF AMERICAN RAILROADS

Rules Governing the Loading of Commodities on Open Top Cars.

(Application for copies should be addressed to the Association of American Railroads, 59 East Van Buren Street, Chicago, IL 60605.)

POWER CRANE AND SHOVEL ASSOCIATION (PCSA)

Standard No. 1 - Mobile Power Crane and Excavator Standards.

(Application for copies should be addressed to the Power Crane and Shovel Association, Marine Plaza, Suite 1700, 111 East Wisconsin Avenue, Milwaukee, WI 53202.)

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SOCIETY OF AUTOMOTIVE ENGINEERS, INC., (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. This specification covers a commercial crane consisting essentially of a crawler mounting, full revolving superstructure, and front-end attachments. Each crane shall be complete and ready for operation. Each crane shall be capable of being converted for use in lifting, clamshell, dragline piledriving, wrecking ball, shovel, and backhoe operations. The crane shall be constructed so that conversion to any one of these uses can be effected in the field without the use of special tools or the necessity of installing additional shafts, drums, clutches or other parts that should for reasons of practicality be included at time of manufacture. The crane shall be capable of traveling over rough ground under adverse or severe conditions. Except for modifications and special equipment required to conform to this specification, the crane shall be the current commercial production model of a standard product and, except for product improvements, shall have demonstrated industry acceptability by having been manufactured, marketed, and sold to commercial users for at least 1 year prior to the opening of bids. The crane shall meet the EPA emissions standards and Power Crane and Shovel Association Std No. 1.

3.1.1 Compatibility. The crane shall be compatible with front end attachments as specified in 3.17.

3.2 First article. The contractor shall furnish one or more cranes for examination prior to starting production to prove that his production methods and choice of design or detail will produce cranes that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and all examinations and tests shall be conducted by the contractor and subject to surveillance and approval by the Government.

3.3 Technical Information Package (TIP). The TIP shall be submitted with all proposals and bids. The TIP shall be completed depicting the crane and its components exactly as it will be furnished. Notation shall be made of all components which have not been in field use for a minimum of 1 year, and shall include identification of the component or part previously used. The manufacturer shall certify, by signing the TIP, that the item to be furnished is identical to those being procured for commercial use, except for modifications permitted in Section 3 of the specification.

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3.4 Material. Materials used shall be free of defects which would adversely affect the performance of components or of the overall assembly. All materials shall be selected by the contractor and shall be of the same quality used for the intended purpose in commercial practice (see 6.8). Unless protected against electrolytic corrosion, dissimilar metals as defined in MIL-STD-889 shall not be used in contact with each other.

3.5 Identical items. All cranes furnished shall be physically and mechanically identical in all respects except with prior written approval of the contracting officer.

3.6 Safety and human factors.

3.6.1 Safety. All belts, gears, shafts, pulleys, sprockets, spindles, drums, flywheels, chains, and other reciprocating, rotating, or moving parts of equipment shall be guarded when such parts may be inadvertently contacted by operating and maintenance personnel or otherwise create a hazard. All hot surfaces of equipment, including exhaust pipes and other lines which may be subject to high temperatures, exposed to contact by personnel or which create a fire hazard shall be fully guarded or insulated. Fuel tanks shall be located in a manner which will not allow spills or overflows to run onto engine, exhaust, or electrical equipment. Exhaust or discharges from equipment shall be directed so that they do not endanger personnel or obstruct the view of the operator. The size and location of access/egress passages shall be in accordance with SAE J833 and SAE J185. The crane shall be equipped with seatbelts designed in accordance with SAE J386. All points requiring lubrication during operation shall have fittings located or guarded so as to be accessible without hazardous exposure to personnel performing these operations. Electrical equipment shall be effectively guarded and grounded to protect all persons and objects from electrical shock hazard. Nonfunctional sharp edges, projecting points, and excessive length of fastening devices shall be avoided. The principal platform walking surfaces shall be of an antiskid type. Engine cooling fans shall have heavy grille or ring type guards. The crane shall meet the applicable requirements of the Occupational Safety and Health Administration and ANSI B30.5.

3.6.2 Human factors. The characteristics of the crane shall provide for operation and maintenance by personnel ranging from the 5th to 95th percentile male and female operator as defined in SAE J833 suitably dressed for the environments specified in 3.10.1.

3.7 Sound Level. The crane shall be equipped with all available standard and optional sound suppression equipment. If the noise level exceeds the 85 dB(A) limitation, the contractor shall provide warning decals in accordance with MIL-STD-1474. If the sound level with cab doors open exceeds 85 dB(A) when it is less than 85 dB(A) with the doors closed, a decal shall be provided stating:

"HEARING PROTECTION REQUIRED WHILE OPERATING WITH DOORS OPEN."

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3.8 Transportability. The crane shall be so designed and constructed that it can be conveniently disassembled into major components that come within the size and weight limitations for rail, sea, and highway transport. The gross vehicle weight of the crane shall be stenciled on each side of the crane in such a manner as to be readily discernable to handling personnel. The crane shall be capable of disassembly in not more than 36 manhours into units of a size that when loaded on a flatcar in accordance with Association of American Railroads Rules Governing the Loading of Commodities on Open-Top Cars, will meet the Berne clearance dimensions as shown in figure 1. The length of each unit shall not exceed the item length shown in figure 1. Reassembly shall require not more than 36 manhours. Cutting and welding are not permitted.

3.8.1 Lifting and tiedown provisions. The crane and its attachments shall be furnished with lifting and tiedown provisions conforming to MIL-STD-209, type III, to facilitate dissassembly and transport. The provisions shall be either integral or permanently affixed attachments.

3.8.2 Antirotation devices for shipping. Steel rods shall be furnished for securing the revolving superstructure to the track frame to prevent rotation during shipment. One end of each rod shall be permanently affixed to the track frame. Means shall be furnished to secure the loose end of the rod to the superstructure for use as an antirotation device and to the track frame for storage.

3.9 Maintainability. The crane design, together with component and accessory location and installation, shall permit ready accessibility in accordance with SAE J925, to all items requiring periodic maintenance or service in the field. Covers, safety guards, and plates which must be removed for component adjustment or servicing shall be equipped with quick-disconnect fastenings. Ease of maintenance provisions shall incorporate features assuring clearances for tools and servicing at temperatures specified in 3.10.1. All maintenance, assembly or disassembly operations required shall be accomplished using common tools and special tools furnished with the crane.

3.9.1 Filling, draining, and checking provisions. The transmission housings and other enclosures, such as gearcases, chain cases and reservoirs which contain lubricant or hydraulic oil shall be equipped with provisions to determine the fluid level. Dipsticks shall be etched, knurled, sand blasted, or phosphate coated to facilitate determining the liquid level. Each enclosure shall be equipped with means for filling and draining. The drains shall be fitted with magnetic drain plugs. Each drain plug or valve shall be located so that removal of the plug or valve will result in complete drainage of the fluid from the enclosure (except torque converter). When the crane is in a level position, drainage of fluids shall be into a 2-gallon bucket. Accessibility to the filling means, the fluid-level checking device, and the drain plug or valve shall be provided without removal or adjustment of accessories and parts, except for plates equipped with hand-operable, quick-disconnect fasteners, or capscrews in applications where quick-disconnect fasteners are subject to damage. The drain outlets for the engine-cooling system shall provide complete drainage of the system. The radiator drain shall be such that coolant will not drain onto any part of the crane. Extension hoses are permitted.

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3.10 Environmental requirements.

3.10.1 Operating temperature requirements. The crane shall perform as specified herein in any ambient temperature from -25° F to +120° F.

3.10.2 Elevation conditions. The crane shall perform as specified herein at any elevation from sea level (barometric pressure 29.92 inches mercury) and a maximum ambient temperature of 120° F to 5,000 feet above sea level (barometric pressure 24.89 inches mercury) and a temperature of 107° F.

3.11 Performance.

3.11.1 Capacities.

3.11.1.1 Crane capacity. The crane, when used as a lifting crane, shall have rated lifting capacities through 360° rotation not less than those specified in table I.

TABLE I. Rated crane lifting capacities.

Boom length (ft)	Radius of load (ft)	Rated load (lb)
40	12	80,000
60	50	10,000

3.11.1.2 Shovel capacity. The crane, when equipped with a shovel attachment (see 6.2), shall have a shovel dipper capacity not less than that specified in table II. The working heights, radii, and digging depth of the shovel shall be not less than those specified in table II with the crane shovel resting on firm level ground and the shovel boom positioned at a 45 degree angle.

TABLE II. Shovel operating range.

Shovel dipper capacity	Dumping				Cutting				Digging depth below floor level		Cleanup radius at floor level	
	Height door open		Radius max. height		Height		Radius					
(cu yd)	(ft)	(in)	(ft)	(in)	(ft)	(in)	(ft)	(in)	(ft)	(in)	(ft)	(in)
2	17	6	28	0	27	0	32	0	7	0	21	0

3.11.1.3 Backhoe capacity. The crane, when equipped with a backhoe attachment (see 6.2), shall hava a capacity, bucket width, ground clearance at start of dump and the digging depth of not less than those specified in table III with the crane shovel resting on firm level ground.

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TABLE III. Backhoe working ranges.

Capacity	Ground clearance at start of dump		Digging depth		Bucket width w/o side cutters (min)
(cu yd)	(ft)	(in)	(ft)	(in)	(in)
2	12	0	25	0	50

3.11.1.4 Clamshell and dragline capacity. The crane shovel equipped with the basic boom shall be adaptable to and operable with a 2-yard capacity clamshell bucket and with a 2-yard capacity dragline bucket.

3.11.1.5 Crane capacity plate. The crane shovel shall be provided with a crane-capacity plate mounted inside the cab in a location conspicuous to the operator. The plate shall contain the following information:

(a) Rated loads in 5-foot increments for the full range of operating radii, angles and superstructure positions when using the basic boom length specified in table I and for each 10-foot increment in boom lengths up to and including 160 feet. This plate shall also include capacity information for jib lengths as furnished.
(b) Indication that the rated loads are loads at a radius measured from the center of rotation of the revolving superstructure to the centerline of the hookblock and that the weights of all lifting means except the hoist rope are to be included as part of the load.
(c) Statement that the rated loads are based on the crane standing on a supporting surface that is firm, level, and uniform.
(d) Statement of deduction to be made from the rated loads for components of crane attachments such as jibs.

3.11.2 Operating characteristics. All operational functions for a given front-end attachment shall operate independently and simultaneously for maximum performance.

3.11.2.1 Maneuverability. The crane shall be capable of traveling in a forward or backward direction with the revolving superstructure in any position, and capable of being steered either to the right or left with controls from the operator's position. The completely equipped crane with applicable front-end attachment located either to the front or back of the crane shall conform to the following mobility requirements:

(a) Capability of ascending, descending, and stopping on a 30 percent grade not less than 35 feet in length on smooth, firm, dry ground.
(b) Capability of negotiating a 15 percent side slope on smooth, firm, dry ground.
(c) Capability of travel on firm, level, dry ground with a maximum speed of not less than 0.75 miles per hour.

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(d) Ground clearance not less than 12 inches.
(e) Capability of fording and operation on hard-bottomed bodies of water up to 12 inches without evidence of damage or water leakage.

3.11.2.2 Ground bearing pressure. The ground bearing pressure of the crane with basic boom, as calculated in 4.4.2, shall be not greater than 15 pounds per square inch (psi).

3.12 Backward stability. The center of gravity from the axis of rotation shall be not farther than 70 percent of the radial distance from the axis of rotation to the tipping fulcrum in the least stable direction when the crane is resting on firm, level, and uneven supporting surface; in working order without load; and with the basic boom at the minimum working radius.

3.13 Crane load stability. Crane stability under loads shall be in accordance with SAE J765. The load rating chart load capacity shall be not more than 75 percent of the stability established.

3.14 Stress requirements. Design of the completely assembled crane shall be such that the maximum stress imposed will not exceed the strength margins contained in SAE J987. Tests in accordance with SAE J987 shall have been performed on the model crane being provided, and test results shall be made available for Government inspection at the time of first article tests.

3.15 Superstructure. The superstructure shall consist essentially of a weathertight cab enclosing the operating machinery and operator's position, mounted on a full revolving platform. The principal members of the platform shall be designed to receive the front-end attachments with mounting pins. Means shall be provided for retaining the pins. The pins shall be capable of being aligned with common handtools. The machinery deck shall be sealed with boots or grommets where controls pass through the deck. Two clips shall be furnished and fastened on the front edge of the operating frame to provide connection points for the inner end of the piledriver lead catwalk (see figure 2). Access systems shall conform to SAE J185. Minimum access dimensions shall conform to SAE J925. A ladder shall be attached to the superstructure cab to facilitate accessibility to the gantry and A-frame sheaves. The fuel tank filler neck shall be accessible and no more than 36 inches from the ground or from an antiskid platform space not less than 15 inches square to enable the operator to fuel the crane tank with 5-gallon cans or to aid in bulk fueling. A catwalk conforming to SAE J185 shall be installed around the superstructure except for the boom foot area. The catwalk shall be bolted to the superstructure and shall be removable. Access footholds or handholds to enable the operator to gain access to all platforms and catwalks shall be provided. Removable cast iron counterweights shall be provided. The complete counterweight shall be capable of removal and placement on a trailer or truck in not more than 2 hours by two men using only common hand tools and the lifting provisions of the crane. The complete counterweight shall be capable of replacement on the crane by two men in not more than 2 hours using the same tools and the lifting provisions of the crane. The counterweight may serve as the rear catwalk providing the walking surface is an antiskid surface and otherwise meets all requirements specified herein.

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3.15.1 Functional requirements. With the engine operating at governed speed, the crane superstructure shall meet the following requirements:

(a) Hoisting-drum single line power of not less than 3,500,000 foot-pounds (ft-lb) per minute with bare drum (single layer) when tested in accordance with SAE J820.
(b) A swing speed of not less than 3.5 revolutions per minute (rpm) at governed engine speed.
(c) Hoisting-drum single line speed between 147 and 180 feet per minute (fpm) at governed engine speed.

3.15.2 Forward machinery deck protection. A sheet metal trough shall be provided to protect the forward machinery deck from mud and dirt scraped from the haul-in cable during dragline operations. The trough shall not interfere with any of the specified operations.

3.15.3 Cab. The cab shall completely enclose the operator's seat, control levers, and pedals, instrument panel, and all main-deck machinery in accordance with SAE J154a. Hinged covers with slots for hoist cables shall be provided for weather protection of the cable drums. The arrangement of windows and windshield shall provide visibility and ventilation for the operator. The seat, instruments, and controls shall be in accordance with SAE J833 and SAE J898. The cab shall have sufficient windows and glazed doors to permit 180-degree vision for the operator. The work area, including the boom point sheaves at the minimum operating radius, shall be visible to an operator varying from large to small in accordance with SAE J833. All glazing shall be tempered safety plate glass. The doors shall be of the sliding type with locks that are accessible. The doors shall be constructed of metal. All doors shall be held open and closed with a positive locking device and shall be provided with latches to prevent accidental opening and closing. The seat, conforming to SAE J899 shall be located and adjustable so that an operator ranging in sizes as indicated in SAE J833 is within reach of the control levers and pedals. Walking surfaces and steps shall be of the antiskid type. Drainage provisions shall be made for any wells or depressions that are subject to collection of water.

3.15.3.1 Cab heating and defrosting. The cab shall be equipped with a commercial heavy duty heater and defroster. The heater and defroster shall be mounted and positioned to permit routine maintenance and inspection and to allow for removal. A ducting system shall be provided to direct heat toward the operator's feet and to the defrosters, and the ducting system shall include valves to permit the operator to select air distribution. Ducting shall not interfere with operator performance and with ease of maintenance, and shall be protected from damage by personnel. Controls shall be located so as to be within reach of the operator in his normal operating position. The defroster shall be so arranged to defrost not less than 75 percent of the front window area.

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3.15.3.2 Windshield wiper. Windshield wiper(s) shall be provided. Wiper motors shall be furnished with not less than a 2 speed control. The wiper arms and blades shall be of the commercial, heavy-duty type having saddle and pin blade fastening. The wipers shall have a park position for holding the blade when the glass is stored overhead. The motor controls shall be mounted on the instrument panel.

3.15.3.3 Superstructure cab, disassembly. For transportability, the superstructure cab shall be capable of removal to reduce the size, weight, and shape of the cross section of the crane. The framing shall tie the side panels and main portion of the roof together as one unit. The top corners shall be provided with lifting eyes to accommodate crane slings. The supporting members of the cab shall be bolted to the machinery deck with fasteners of sufficient strength and quantity so as to hold the cab securely in place under all operating conditions. All free-standing edges of the removable and fixed section shall be provided with stiffeners and self-sealing, labyrinth-type weather seals to keep out water. All electrical, air, and hydraulic lines fastened to the walls of the cab shall be equipped with quick-disconnect, self-sealing couplings as needed at the separation line of the cab and the machinery deck.

3.15.4 A-frame. The A-frame or gantry provided shall be capable of withstanding stresses induced by lifting a 160 foot boom from the ground to operating position. The principal vertical compression and tension members of the A-frame shall be connected so that the operating stresses are transferred to the main structural members of the rotating frame of the superstructure. The gantry shall be of the folding or retractable type for overhead clearance or reducing to cube during transportation.

3.15.5 Power load-lowering. For crane operations, the primary load hoisting drum shall be equipped with a powered reversing drive and suitable control features to provide precision controlled power lowering of loads by the main hook. The powered load lowering mechanism shall be controlled by a lever conveniently located within the revolving superstructure, and shall be capable of lowering all rated loads with the manufacturer's recommended reeving. The powered load lowering mechanism shall be capable of lowering the loads at a smooth, steady controlled rate of speed; and a smooth variable controlled rate of speed; dependent on engine speed, without jerking, bouncing, or slipping the load. A braking device may be used to assist the power lowering mechanism. Extended use of the powered load lowering mechanism shall not result in clutch or brake failure by fading, grabbing, cracking, internal rupture, breakage, permanent deformation, or any malfunction. Clutch and brake lining temperatures shall not exceed the maximum recommended by the manufacturer of the lining. The powered load lowering mechanism shall be capable of being disengaged, when not in use, so as not to interfere with normal crane operations.

3.15.6 Crane-boom hoist. The crane-boom hoist shall be independently controlled and shall safely suspend the boom with recommended reeving without attention from the operator, shall provide for controlled lowering under the operator's control, and shall make free spooling of the boom cable impossible

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under any circumstance. The boom hoist shall raise and lower the boom with rated capacity loads. It shall have sufficient rope capacity to allow the boom tip to be lowered to ground level with the crane resting on level ground and not less than three wraps of wire rope retained on the drum. The boom hoist shall be equipped with a positive lock to facilitate holding the boom in any set position. The boom-hoist drum shall be arranged to minimize eccentric pull in order to avoid chafing of the rope and unequal wear on the drum shaft bearings. The drum shall spool the rope in even layers. The boom hoist clutch shall be readily accessible for adjustment and repair.

3.15.7 Drums and sheaves. The drums shall be independently driven and suitable for use with any of the different front-end attachments and arranged so as to reduce eccentric pull to a minimum in order to avoid chafing of the rope and unequal wear on the drum shaft bearings. The complete drum assembly and mounting shall rotate freely when clutches and brakes are not engaged to allow gravity lowering of the hook block. The drum shall be equipped with a socket and wedge for dead-ending the wire rope. Lubricating fittings shall be located to permit lubrication without unspooling cable. Drums shall have sufficient rope capacity to accommodate the required rope lengths. Pitch diameters of drum and drum laggings including the boom hoist drum shall be in accordance with SAE J881. The machinery operating the primary and secondary drums shall be arranged so that with the engine-clutch disengaged the drum clutches may be simultaneously engaged, to reverse the operating machinery by lowering the load on either drum under control of both drum brakes simultaneously for a precision load lowering without the boom being raised. All drums shall be guarded to prevent the rope from leaving the drum during operation. The pitch diameter of load bearing sheaves shall be in accordance with SAE J881. Flanges on the boom point sheaves shall extend beyond the rope grooves not less than 1.75 times the diameter of the rope. All sheaves shall be guarded to prevent the rope from leaving the sheave groove during operation. The sheave pins shall be of sufficient size and strength to support the sheaves under maximum operating conditions.

3.15.7.1 Drum brakes. The drum brakes shall hold and slowly lower without jerking or chattering, capacity loads supported by recommended reeving. Provision shall be made for locking the brakes in any position of the drum. The drum brake shoes shall be easily removable for lining replacement.

3.15.7.2 Drum clutches. Drum clutches shall be capable of withstanding rated speed and continuous operation without fading, accessible for adjustment and repair, and sufficiently large to eliminate slippage under 133 percent of rated load with recommended reeving. The primary and secondary load hoist drums shall be operated by clutches controlled by levers and shall be designed so that once the clutch is engaged, it shall be held in place to hold the clutches in engagement and be able to perform the power lowering tests specified in 4.4.9. The clutches shall be unlocked by pulling or pushing forward on the control lever.

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3.15.7.3 Auxiliary drum. When specified (see 6.2), an auxiliary or third drum shall be provided and shall meet the requirements of 3.15.7, 3.15.7.1, and 3.15.7.2. The drum shall be engaged by a clutch operable from a control lever in a position within easy reach of the operator while seated.

3.15.8 Rollers. The rotating frame shall be supported by fixed rollers, or by a live ball or roller ring. Rollers, balls, roller rings, and ball races shall be heat-treated hardened steel and shall be accurately machined. Fixed rollers shall be fitted with antifriction bearings and shall be provided with grease fittings. The balls or rollers of live ball or roller rings shall be accessible and shall be removable. When the hold-down function is performed by hook rollers, such rollers shall be either the dual-purpose type in which each set of rollers performs both the supporting and hold-down functions, or the rollers may be of the hold-down type only. Hook rollers shall operate either singly or in pairs on tandem bogies and shall be provided with grease fittings. Hook rollers shall be provided with means for vertical or horizontal adjustment as required. The hook rollers shall be removable without removal of the hook-roller brackets from the revolving superstructure. The hook roller brackets shall not interfere with the removal of the superstructure from roller path and frame. The temperature of the supporting balls or rollers and hook rollers shall not exceed 175° F when tested in accordance with 4.4.7.

3.15.9 Swing mechanism. The rotation of the rotating frame shall be controlled through clutches designed for fast cycling, craning, and other operations, and capable of transmitting not less than 75 percent of the maximum engine torque. The temperature of the swing clutches shall not exceed the maximum temperature recommended by the clutch lining manufacturer. The swing speed at governed engine speed shall be not less than 3.5 rpm.

3.15.9.1 Swing-gear and lock and brake mechanisms. The revolving superstructure swing shall be actuated by a gear and pinion. A swing lock shall be provided to lock the revolving superstructure in any desired position. A swing brake shall also be provided to control the swing of the revolving superstructure and to hold the superstructure from swinging when the crane is working on a slope and shall require manual effort to engage. Both lock and swing-brake controls shall be located within reach of the operator while seated.

3.15.10 Power transmission. The engine shall be connected to the transmission through a friction disengaging clutch and a hydraulic torque converter. The clutch and torque converter shall be located so as to be accessible for adjustment and repair.

3.15.10.1 Clutch. The clutch shall be controlled from the operator's position and shall be capable of being engaged and disengaged instantly while the engine is running.

3.15.10.2 Drive chains. All drive chains shall be provided with a means of adjustment. This adjustment shall be made either by changing the center to center distance between the driving and driven shafts or by means of a roller or sprocket meshing with or riding on the chain. This roller or sprocket shall

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be located so the tensioning action shall be exerted on the exterior side of the slack side of the chain. The primary drive chain between the torque converter output shaft and the first sprocket shall be encased. The case for this chain shall provide a leak-proof lubricant reservoir.

3.15.11 Electrical system. The crane shall be equipped with a 12- or 24-volt, negative grounded, waterproof electrical system. The electrical system shall be in accordance with SAE J821 with circuits identified by the number option. All components shall be provided with the maximum protection from damage, shock, and vibration through means of location and furnishing of resilient mounts and cushions.

3.15.11.1 Electrogmagnetic interference. The electromagnetic interference emission of the crane shall conform to SAE J551 when tested in accordance with 4.4.18.

3.15.11.2 Wiring. The wiring of the crane shall be fabricated and installed to provide a reliable electrical system. Conduit shall be furnished for all circuits to within 12 inches of components, with the exception of the battery cables. Conduit, clips, loom, grommets, and cushioned support clamps shall prevent whipping, chafing, or other mechanical damage to the wiring and harness. Wiring shall be routed within the framing members whenever possible. Removal of major assemblies such as engine, cab, and instrument panel cluster shall be accomplished by disconnecting one or more plugs and receptacle with the exception of battery, starter, and slave cables. All circuits shall be protected by circuit breakers in accordance with SAE J553 or fuses in accordance with SAE J554.

3.15.11.3 Lights. Lights shall be controlled by switches, and be protected by circuit breakers. Switches shall be located on the instrument panel in the revolving superstructure within reach of the operator when seated. One switch and associated electrical circuit shall be provided for the independent control of the following light combinations, as furnished:

(a) Revolving superstructure floodlights.
(b) Interior cab lights.
(c) Instrument lights.

3.15.11.3.1 Floodlights. Not less than six floodlights, conforming to SAE J598 shall be furnished and shall be attached to the crane on mounting brackets providing vertical and horizontal adjustments. The housings shall be mounted in a position to permit replacement of the lamp. Lamp housings shall not extend beyond the side of the revolving superstructure unless provision is made for folding the lamps clear of the sides of the superstructure for transport. The floodlights shall be located so that lenses will not shine directly into the operator's eyes when the operator is working in the normal operating position. The floodlights shall adequately illuminate the working area so that typical work may be performed at night. Not less than four floodlights, two on each side of the boom hoist cables, shall be mounted at the top front edge of the cab

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to illuminate the working area. Not less than two floodlights, one on each side of the boom mounting shall be mounted as near the side and bottom of the superstructure as practicable.

3.15.11.3.2 Interior lights. The cab shall have not less than one interior ceiling light with plastic lens not less than 4 inches in diameter. The light(s) shall be effectively located. Each light shall be equipped with a bayonet type bulb. The overall depth of the light fixture from the face of the lens to the ceiling shall not exceed 3 inches.

3.15.11.3.3 Trouble lamp. A handgrip type, portable, trouble light shall be mounted in the cab adjacent to the operator's door. The trouble light shall have a handle incorporating an on-off switch. The trouble-light mounting bracket shall protect the sealed beam unit from damage.

3.15.11.4 Warning horn. The revolving superstructure shall be equipped with a warning horn. The horn shall be mounted in a protected location on the exterior of the revolving superstructure. The horn shall be operated by a pushbutton switch mounted within reach of the operator.

3.15.11.5 Instrument panel. An instrument panel containing all switches, gages and instruments, except the hourmeter which may be mounted on the engine, for complete control of the engine and lights, shall be located in accordance with SAE J209, and be placed within reach and visibility of the operator in his normal operating position. The instrument cluster group shall be removable to permit access to all wiring and units. Panel lights shall be furnished for visual reading of all instrument, indicators, and identification plates by the operator. Each switch, control, or gage on the instrument panel shall be labeled in accordance with SAE J298. Arrows or any appropriate mark showing the direction to operate, if pertinent, shall be provided. Identification plates furnished shall conform to 3.19. The tachometer shall be accurate within plus or minus 5 percent of actual engine speed.

3.15.11.6 Starting systems. The engine starting systems shall be as recommended by the engine and starter manufacturers for this application. The starting systems shall include a starter switch, engine speed disconnect, main starter solenoid, starter motor and flywheel drive mechanism.

3.15.11.7 Charging system. An engine-driven battery-charging system shall be furnished. Adjustable generator or alternator mounting brackets shall provide movement for adjusting belt tension, and shall resist belt tension forces to minimize potential loss of recommended belt tension.

3.15.11.8 Fungus and moisture resistance. The electrical circuitry, including all components and connections shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173.

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(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

The varnish shall be applied to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.15.11.9 Storage batteries. Storage batteries of adequate capacity for the application and in accordance with SAE J930 shall be furnished. The electrolyte for each crane shall be separately packaged in accordance with 5.1. The positive and negative terminals shall be identified.

3.15.11.10 Battery container. A battery container shall be provided for the batteries. The container shall be corrosive-resistant and shall provide for (1) holding the batteries level when the crane is on level ground, (2) drainage for water or electrolyte spilled within the enclosure, (3) gas venting to the outside, at or near the top of the enclosure, and (4) accessibility for servicing and removal of the batteries. Clear vision shall be provided for checking of the electrolyte level in each cell of the batteries.

3.15.12 Engine. The commercial diesel engine shall be a heavy duty industrial type, capable of operating on diesel fuel conforming to VV-F-800 and lubricating oil conforming to MIL-L-2104. The diesel engine shall start within 5 minutes and be ready for full load operation within 15 minutes in any ambient temperature from -25° F to +120° F. When a fluid priming system is required, it shall be of the measured shot type with storage capacity of at least 12 fluid ounces.

3.15.12.1 Oil sampling valve. An oil sampling valve shall be provided on the engine and shall close automatically when released. It shall be made of material resistant to corrosion and shall not contaminate the sample. The valve shall be located so that personnel will not be exposed to danger when taking oil samples with the engine running. The location of the oil tap shall be such that when samples are taken, it shall be a true representation of oil that is flowing while the engine is running.

3.15.12.2 Operation conditions. All systems, including fuel system, lubrication system, air induction system, and coolant system, shall perform without leakage or malfunction when the crane is negotiating a 30 percent grade or a 15 percent side slope.

3.15.12.3 Fuel system. The capacity of the fuel tank shall be sufficient for 8 hours of continuous operation at 75 percent of the rated intermittent brake horsepower. The tank shall be equipped with a drain outlet and threaded plug. The drain outlet shall be located so that removal of the plug will result in complete drainage of the tank, when the crane is on level ground. The fill pipe

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shall be marked "DIESEL FUEL ONLY". The tank filling components shall include fill pipe, strainer, safety cap, and external vent. The fill pipe shall be 4-inch minimum diameter and shall be in a protected and accessible location. The fill pipe shall be equipped with a safety cap.

3.15.12.4 Air induction system. The engine shall be equipped with a dry-type air cleaner. The intake vacuum of the installed system shall not exceed maximum vacuum recommended by the engine manufacturer.

3.15.12.5 Engine exhaust system. An engine exhaust system shall be furnished, and shall expel gases away from the operator's cab.

3.15.12.6 Governor. The engine shall be controlled by a variable speed governor. Selected governor settings through the operating range shall be provided by a control lever which shall be accessible to the operator in the seated operating position.

3.15.13 Operator controls. The crane shall be controlled by the minimum quantity of hand levers and foot pedals necessary for efficient operation. All levers and pedals shall be conveniently located in accordance with SAE J898 and arranged in accordance with SAE J983 for easy access and operation by the operator when seated in the normal operating position without causing obstruction to the operator's view of the work. All brakes shall be capable of being easily locked and unlocked. Control levers shall be free of all latches which may seriously delay emergency operation. All levers and pedals shall be permanently marked as to the function it controls. When the same lever controls more than one function, all such functions shall be marked. These markings shall be made on plates. The plates shall be located adjacent to the control and visible to the operator.

3.16 Crawler mounting.

3.16.1 Crawler base. The crawler base shall be steel casting or shall be rigidly constructed of heavy steel shapes, fitted and welded to prevent permanent distortion when traveling across sharp depressions and uneven ground. The swing gear and turntable roller path shall be steel and shall be rigidly attached to or integral with the base. The turntable roller or shear ball path shall be of sufficient size and strength to support the revolving superstructure under all conditions without distortion. Where hook rollers perform the holddown function, the lower member of the center pivot bearing shall be integral with, shrunk into, or welded to the crawler base. The center pin or gudgeon shall operate in bronze or antifriction bearings.

3.16.2 Track assembly. The crawler tracks shall be constructed for operation in loose dirt, mud, rocks, and snow; shall be provided with suitable adjustments for maintaining correct tension in the track; and shall be constructed so that the track cannot be accidentally disengaged. The track shoes shall be of wear-resisting steel and of flat self-cleaning construction. The track-connecting pins shall be of steel, of sufficient size, hardness, and strength of the purpose intended, and shall be removable and replaceable. The

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tracks shall be supported on a sufficient number of steel rollers in such a manner so as to distribute the weight of the entire crane over the ground for the entire length of the crawlers without depending upon the tension in the track for support. All track assembly shafts and rollers shall be provided with guarded grease fittings. Where practicable, grease fittings shall be located so as to be accessible from outside the crawlers and lower base.

3.16.3 Side frames. The crawler track side frames shall be steel castings or welded structural steel members.

3.16.4 Crawler drive. The crawler shall be driven by gears, roller chains, or hydraulic motors. The travel mechanism shall have positive steering controlled from the operator's cab. The crawler clutches shall be of adequate size to permit constant operation without overheating. Brakes or other locking means shall be provided to hold the crane stationary during the working cycle while the crane is on level grade, and to prevent the crane from accidentally rolling down a 30 percent grade. Brakes that can be applied at any time shall be furnished on the travel mechanism. It shall be possible to apply the brakes on both tracks at the same time. Jaw clutches, when furnished, shall have a sufficient number of teeth to engage and to transmit maximum traction power without undue wear. The travel gears and bearings shall be protected against the entrance of dirt or other foreign matter. The propelling mechanism shall be provided with a positive traction lock or brake for locking the crawlers. The propelling shaft shall be so arranged that it is not one or a part of the supporting axles. All shafts shall be bushed or provided with antifriction bearings and ball-check grease fittings.

3.16.5 Axles. Each axle or crossmember shall be either of cast, forged, fabricated or structural steel.

3.17 Front-end attachments. The revolving superstructure of each crane shall be equipped with all necessary equipment to permit operation as a crane (boom with or without a jib), clamshell, dragline, piledriver, shovel, backhoe, and crane with a wrecking ball. Tools, equipment, parts, and material which are not a component of the crane and are required for operation of front-end attachments shall be furnished with each attachment.

3.17.1 Primary front-end attachments. The following attachments shall be furnished with each crane and such attachments shall be complete with all necessary parts, including wire rope.

3.17.1.1 Crane boom. The crane boom shall be suitable for crane, dragline, clamshell, piledriver, and wrecking ball operation. The basic boom shall be 40 feet in length, and shall consist of two sections. The boom shall be capable of being extended to a maximum length of 160 feet by the use of boom extensions (see 3.17.2.1). The boom point shall be equipped with not less than three guarded parallel sheaves conforming to 3.15.7 and with not less than two independent dead-ends commensurate in size with the wire rope furnished. Piledriver adapter plates conforming to figure 3 shall be provided to adapt the leads specified in 3.17.2.8 to the crane boom. Adapter plate support bracket(s) shall be furnished on the inner section of the boom.

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3.17.1.2 Pendant equalizer assembly. A pendant equalizer assembly shall be furnished.

3.17.1.3 Pendant cable. A matched set of pendant cables shall be provided for connecting the pendant equalizer assembly to the boom point.

3.17.1.4 Boom safety stop. A telescopic boom stop shall be furnished to provide a safeguard against overtopping of the boom. The complete boom stop with supporting attachments shall have sufficient strength to develop the ultimate strength of the boom in bending at the point of attachment of the boom stop to the boom. The boom stop shall provide a shock-absorbing action and shall remain assembled when the boom point is lowered to the ground. The boom safety stop shall conform to SAE J220. Under all operating conditions specified herein, as the boom is raised from the ground, the boom stop shall telescope freely without sticking, binding or bending while the boom is raised to the maximum allowable angle.

3.17.1.5 Boom hoist clutch throwout. Means shall be provided to disengage the clutch of the boom hoist drum when the boom has reached the maximum angle or height and is in contact with the boom stops.

3.17.1.6 Boom angle indicator. An indicator which will show the position of the boom at all times shall be provided. The indicator shall be of the heavy-pendulum type swinging on a double-sealed antifriction bearing. The pointer shall indicate the angle of the boom with the horizontal plane. The indicator shall be located on the boom, near and visible to the operator in his normal operating position. A chart located in the cab near the operator shall be provided showing rated lifting capacities at various boom angles for various boom lengths, with a statement of deductions to be made when using the jib. The boom angle indicator shall conform to SAE J375.

3.17.1.7 Overload indicating system. When specified (see 6.2) a load indicating warning system of the load moment type shall be furnished in accordance with SAE J159, SAE J 375, and SAE J376. This system shall automatically indicate and warn the operator when an overload condition exists that exceeds the rated loads specified in 3.11.1.1 through the use of lights, displays, and an audible signal mounted within the operator's view within the cab. The audible and light signals shall cease and displays shall indicate a safe loading condition when corrective action has been taken by the operator. The indicating system may utilize the electrical system of the crane. The electrical system shall be waterproof and shall perform under all operational conditions specified herein without any malfunction.

3.17.1.8 Crane hook block. A commercial crane hook block with safety latch suitable for the application shall be provided.

3.17.1.9 Wire rope. All wire rope except for the tagline shall be IWRC improved plow steel, preformed right regular lay wire rope conforming to RR-W-410. Wire rope strength factors shall be in accordance with SAE J959.

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3.17.1.9.1 Rope sockets. All wire ropes, except tag lines, pendant cables and dipper trip lines, shall have ends attached to drum laggings or to dead-end connections by wedge-type sockets. Clamps, clips, and screw wedge connectors shall be permitted only on tag line and dipper trip lines. Boom pendant cables specified in 3.17.1.3 shall have swaged open-end sockets, and pendant cables for boom extensions specified in 3.17.2.1 shall have one open-end socket and one closed-end socket.

3.17.1.9.2 Rope lengths. Wire rope shall be of the greatest length required to perform as a crane performing the functions specified in table IV and based on the maximum boom length furnished. The boom hoist shall have sufficient wire rope to allow the tip of the boom to be lowered to rest on level ground with three wraps of wire rope remaining on the boom hoist drum.

TABLE IV. Crane function to determine rope lengths.

Function	Position of boom	Position of attachment
Crane hook	At minimum radius.	At ground level.
Clamshell	60° from horizontal.	20 percent of boom length (in feet) below ground level.
Dragline	40° from horizontal.	50 percent of boom length (in feet) below ground level and 30 percent of boom length (in feet) beyond boom tip.
Piledriving	At basic boom height.	Two basic boom lengths +50 feet below ground level.
Auxiliary drum hoisting	At maximum boom height.	Two boom lengths +20 feet.

3.17.2 Secondary front-end attachments.

3.17.2.1 Boom extensions. Boom extensions shall be furnished to provide the specified boom length (see 6.2). Extensions shall provide for an increase in the boom length in 10-foot increments and shall be furnished with matched sets of pendant cables. Extensions shall match and mate with the other boom sections.

3.17.2.2 Boom jib. When specified (see 6.2), a boom jib shall be furnished complete, and shall be readily attached to the crane boom (without piledriver leads in place). Boom jib lengths shall be as specified (see 6.2). Boom jib shall be provided with necessary riser and holdback ropes, and complete with adequate overhaul ball and hook.

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3.17.2.3 Clamshell bucket. When specified (see 6.2), a clamshell bucket shall be furnished and shall conform to MIL-B-16028, type II. Bucket teeth shall be in accordance with MIL-T-52176. Side and corner teeth shall be furnished.

3.17.2.4 Tag line. When specified (see 6.2), a tag line shall be furnished and shall be of the reel type, with an adjustable tension mechanism, shall be in accordance with MIL-T-52198, and shall control the bucket specified in 3.17.2.3. The tag line shall be provided with means of attaching to the boom without cutting, welding, or drilling of the boom and shall be removeable and replaceable with common tools. All necessary parts for installations shall be furnished.

3.17.2.5 Dragline bucket. When specified (see 6.2), a dragline bucket shall be furnished and shall conform to MIL-B-16029, type II. Bucket teeth shall be in accordance with MIL-T-52176.

3.17.2.6 Dragline fairlead. When specified (see 6.2), a dragline fairlead shall be furnished, shall be of the four-sheave swinging or two-sheave revolving type and mounted in accordance with the manufacturer's standard procedure. The fairlead shall have either replaceable wearing plates or replaceable guide rollers, shall have bronze-bushed or antifriction bearing-mounted sheaves with smooth cable grooves, and shall be arranged to avoid chafing the drag cable in any working position.

3.17.2.7 Wrecking ball. When specified (see 6.2), a wrecking ball shall be furnished and shall conform to MIL-B-23276, size as specified.

3.17.2.8 Piledriver leads and catwalk. When specified (see 6.2), piledriver leads and catwalk shall be furnished and shall conform to MIL-L-11025. Lead sections shall consist of one top section and lower sections in numbers as required (see 6.2).

3.17.2.9 Piledriver hammer. When specified (see 6.2), a piledriver hammer shall be furnished. The piledriver hammer shall conform to MIL-H-21103, size as specified.

3.17.2.10 Shovel-front. When specified (see 6.2), a shovel front shall be furnished, and shall operate in accordance with the requirements of table II.

3.17.2.10.1 Shovel boom. The shovel boom shall be of a sturdy construction, strongly reinforced and braced, and shall be fabricated from structural steel shapes and plates rigidly welded to form a box section up to the dipper shaft for an inside dipper stick and to the boom tip for an outside dipper stick. Replaceable wearing shoes or other suitable fittings shall be provided where the dipper stick passes by or through the boom. Removable and replaceable buffer rails shall be provided where the dipper strikes the boom.

3.17.2.10.2 Dipper stick. The dipper stick shall be of rigid and durable steel construction and shall resist twisting strains.

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3.17.2.10.3 Shovel dipper. The shovel dipper shall be of either heavy steel plate, cast steel, or a combination of both. The front lip shall be of abrasion- and wear-resistant, hard-surfaced steel. The bucket lips shall be equipped with tooth sockets, shanks, and teeth in accordance with MIL-T-52176. The dipper shall have a discharge gate, greater in area than the dipper mouth, in the bottom securely hinged to the back of the dipper with large steel hinges. The dipper front-latch opening on the wearing side shall have a replaceable wear plate where the latch bar engages. The dipper angle shall be adjustable for different digging conditions. A power trip controlled from the operator's seat and a slack takeup to prevent fouling of the trip line shall be provided. The shovel dipper capacity shall be not less than 2 cubic yards, obtained by multiplying the mean height of the dipper by the inside cross-sectional area of the dipper at one-half the minimum height plus or minus 2 percent. The mean height shall be one-half the sum of the maximum and minimum height of the dipper body including its minimum lip projection without teeth.

3.17.2.10.4 Dipper crowd and retract. The dipper crowd shall be self-adjusting at all boom angles and shall be provided with a tension adjustment for maintaining the correct tension in the chain or rope. The dipper retract and crowd shall be operated by an independent positive chain or wire rope mechanism.

3.17.2.10.5 Sheave. The shovel boom point sheaves shall conform to 3.15.7 except that the pitch diameter shall be not less than 32 times the diameter of the rope used.

3.17.2.11 Backhoe. When specified (see 6.2), a backhoe shall be furnished and shall operate in accordance with table III.

3.17.2.11.1 Boom, dipper arm, and gantry. The boom, dipper arm, and gantry shall be made of rigidly welded or riveted steel members.

3.17.2.11.2 Dipper. The backhoe dipper shall be constructed of heavy steel plate or of cast steel; shall have a cutting lip of abrasion- and wear-resistant hard-surfaced steel; shall have the bottom reinforced and protected by the extension of the tooth sockets or adequate wearing bars; shall have teeth in accordance with MIL-T-52176, and shall have replaceable, abrasion- and wear-resistant, heat-treated and tempered, drop-forged steel, or cast alloy steel side cutter.

3.18 Lubrication. Unless otherwise preserved for packaging (see 5.1), all components shall be filled to the operating level with military lubricants. Lubrication fittings shall be in accordance with SAE J534. Pressure lubrication shall not damage seals or other parts. All parts requiring lubrication shall be lubricated prior to delivery and tagged to show the type, temperature rating and grades of lubricants used.

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TABLE V. Service products.

Item	Application	Specification	Type & Grade
1	Engine:	MIL-L-2104	10, 30
2	Fuel: Diesel	VV-F-800	DF-1, DF-2, DF-A
3	Grease: Automotive	MIL-G-10924	One grade
4	Hydraulic-system oil: (excl. brakes) Oil	MIL-L-2104	10
5	Torque converter - transmission	MIL-L-2104	10
6	Lubricating oils: Universal	MIL-L-2105	75, 80, 90
7	Antifreeze: Ethylene glycol	MIL-A-46153	One grade
8	Brake fluid	MIL-B-46176	10

3.18.1 Lubrication fittings. Lubrication fittings shall be located in a protected position and shall be accessible to a grease gun conforming to MIL-G-3859 with a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Engine-housing side panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility to lubrication fittings.

3.19 Identification plates. The crane shall be provided with a plate or decal containing the following information:

Nomenclature
NSN
Serial Number
Model
Contract Number
Shipping Date
USA Number

The plate shall be furnished by the manufacturer and permanently affixed. Identification plates or decals shall also be securely fastened to each crane

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attachment furnished, including the boom, jib, and each crane boom section in a conspicuous location. The crane attachment identification plates shall include the following information.

(Attachment nomenclature)

FOR USE WITH

(Name of crane shovel manufacturer(s))

(Model number or numbers of crane)

(NSN)

*(Range of applicable crane serial numbers)

Weight

(Center of gravity location)

*The range of crane serial numbers will encompass the lowest machine serial number and the highest known machine serial number designed to use the attachment without modification. When an attachment is common to more than one crane model number, all applicable model numbers shall be listed.

3.19.1 Instruction plates or decals. Instruction plates or decals shall be furnished and permanently affixed by the contractor in a conspicuous and protected location on the crane. The plates shall include instructions, warnings, cautions, and special or important procedures to be followed in operating or servicing the crane. Plates for transportation data shall show the silhouette of the crane in transport position, indicating locations and capacity of lifting and tiedown attachments, location of center of gravity, length and size of slings required, and the recommended tiedown arrangement for the crane shovel. The lettering shall be not less than 1/4 inch in height. The contractor shall install the electrical powerline warning decal set and vehicle weight classification sign kit furnished by the contracting officer (see 6.4).

3.20 Treatment and painting. All parts of the crane normally painted shall be cleaned, treated, and painted in accordance with good commercial practice. The primer and finish coats, the dry film thickness and the forest green camouflage color shall be in accordance with MIL-T-704.

3.21 Stenciling. Lifting and tiedown attachments shall be identified with 1-inch letters indicating "sling point" or "tiedown". Unless otherwise specified herein, stenciling shall conform to MIL-T-704. Registration numbers shall be marked on the crane in accordance with MIL-STD-642. The gross weight of each crane with equipment shall be stenciled on each side of the frame in such a manner as to be discernable by dock and military personnel.

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3.22 Toolbox. A watertight, lockable toolbox made of steel shall be attached to the crane. The toolbox shall be sufficiently large to contain all the tools required for the routine maintenance and operation of the crane. The toolbox shall be securely mounted in an accessible, protected location permitting the lid to be fully opened. The lid shall be held fully open by the weight of the lid or a suitable fastener. Drainage of the tool box shall be provided.

3.23 Government furnished equipment (GFE). The following will be furnished by the Government for use under the contract:

(a) Supplemental Operating, Maintenance and Repair Parts Instructions (SOMARPI) (when required).
(b) Required maintenance forms.
(c) Initial Prescribed Load List and Initial Authorized Stockage List.
(d) Electrical Warning Decal Set, NSN 7690-00-267-5778.
(e) Vehicle weight classification sign kit, NSN 9905-00-565-6267.

3.24 Government-loaned property. Unless otherwise specified (see 6.2), the following equipment in the quantities indicated will be loaned by the Government for the first article demonstration.

Item No.	Description	Identification (FSN)	Quantity
1.	Lead Section, top, Pile Driver: 15 Ft. Lg.	3815-983-8029	1
2.	Lead Section, Lower, Pile Driver: 10 Ft. Lg.	3815-221-2215	4
3.	Catwalk, Pile Driver, Telescoping: 3 Sect; 8 to 23 Ft.	3815-190-3308	1
4.	Hammer, Pile Driver, Self-Powered: Dsl. Drvn. 8,000 Ft. Lb., Size D	3895-769-5308	1

3.25 Servicing and adjustment. The contractor shall service and adjust each crane for operational use upon delivery.

3.26 Workmanship. All parts, components, and assemblies of the crane shall be cleaned and free of any harmful extraneous material. External surfaces shall be free of burrs, sharp edges, and corners except when sharp edges or corners are required.

3.26.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

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3.26.1.1 Castings and forgings. All castings and forgings shall be clean of harmful extraneous material such as sand, dirt, sprues, scale, and flux. Rework shall not reduce typical mechanical properties or affect a castings function.

3.26.2 Bolted connections. Boltholes shall be accurately formed or drilled and all burrs and nonfunctional sharp edges shall be removed. All bolts, nuts, and screws shall be tight. Adjoining surfaces shall be flat, and in full contact with each other. Lockwashers or other locking means shall be provided for all bolts, nuts, and screws.

3.26.3 Riveted connections. Rivet holes shall be accurately formed or drilled and shall have the burrs removed. Rivets shall completely fill the holes. Rivet heads shall be concentric with the rivet, and shall make full contact with the intended surface. The rivets shall be so fabricated as to develop a joint strength not less than the design value.

3.26.4 Welders and welding. Welding procedures and welder qualification shall be in accordance with ANSI B30.5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Examination. Each crane shall be examined for compliance with the requirements specified in Section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements, shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirement or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.3 First article inspection. First article inspection shall be performed. This inspection shall include the examination of 4.2 and the tests of 4.4. The first article may be a standard production item from the contractor's current inventory, provided the crane meets the requirements of this specification and is representative of the design, construction, and manufacturing technique applicable to the remaining cranes to be furnished under the contract. The contracting officer should include specific instructions in all acquisition documents regarding arrangement for examinations, tests, and approval of the first article (see 6.2).

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4.4 First article tests. The first article shall be subject to the tests specified below. Failure to pass any test, evidence of any leakage distortion, deformation or excessive wear shall be cause for rejection.

4.4.1 Maneuverability test. The crane shall be given the tests necessary to determine that its maneuverability conforms to that specified in 3.11.2.1.

4.4.2 Ground bearing pressure test. The ground bearing pressure shall be calculated by dividing the total working weight with basic 40 foot boom, but not including load, by the bearing area in square inches. The weight shall be determined by platform scales, by calibrated hydraulic, or electronic devices and the bearing area used shall be the center-to-center distance between crawler end sprockets multiplied by twice the width of the track shoe.

4.4.3 Stability test. Crane stability test shall be conducted in accordance with the procedures in SAE J765.

4.4.4 Crane hook test. The crane manufacturer shall furnish certification, from the crane hook manufacturer, that the hook furnished with the crane is without defects or discontinuities and is of compatible size, having the proper load capacity and safety factor.

4.4.5 Lifting capacity tests. Lifting capacities of crane shall be demonstrated by use of actual weights using the minimum (40 foot), maximum (160 foot), and one intermediate length boom. The loads at the tipping point and the boom radius shall be measured as specified in SAE J765 for 3 radii at each boom length.

4.4.6 Power and speed test. Power and speed test shall be conducted in accordance with SAE J820.

4.4.7 Swing test.

(a) With the 40 foot boom at minimum radius, rotate the superstructure at full throttle for 10 minutes in each direction in order to determine if excessive heat is generated in the rollers in conformance to 3.15.8.
(b) With the 40 foot boom at minimum radius, continually oscillate the superstructure at full throttle through an arc of 180 degrees for 10 minutes, to determine if excessive heat is generated in swing clutches.
(c) Test the swing lock or brake at an engine speed of not less than 200 rpm above low idle speed by actuating it in several positions and attempting to swing the superstructure from those positions.
(d) Test for a swing speed conforming to 3.15.1.

4.4.8 Travel brake and lock test. Test the service brakes or other locking means to demonstrate conformance to 3.16.4.

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4.4.9 Powered load lowering test for crane. The powered load lowering system shall be tested as follows to determine conformance to 3.15.5:

(a) With the crane equipped with boom, position the revolving superstructure in the most stable craning position and assemble a single line load equal to the maximum rated load divided by the recommended number of parts of hoist line reeving.
(b) Position the boom angle for the minimum working radius.
(c) Hoist the load 20 feet above the ground level, ± 3 inches, or until interference with the crane boom is obvious.
(d) Prepare the crane for power-down operation on the main load line.
(e) Lower the load under power. Vary the speed of the load by varying the speed of the engine.
(f) When the load reaches the ground, immediately repeat the operations (c), (d), and (e). These operations shall be repeated until 20 cycles have been completed.
(g) Check the temperatures of all linings employed in this operation with at least four thermocouples imbedded within each lining, near the working face, and uniformly spaced around the perimeter, or with a contact pyrometer placed directly on the lining, or on the drum surface near where the lining contacts the drum.

4.4.10 Load holding test. Suspend load; stop the engine; positively lock the main drive mechanism, positively apply the primary hoist drum clutch and release the brake holding the load. As the brake is released, inspect the clutch for slippage. The clutch shall have adequate capacity to hold this load without slipping. Repeat this test using the secondary hoist drum clutch. Suspend a capacity load; stop the engine; positively engage primary drum clutches; disengage the master clutch and lower the load on the brake, backing down the hoist machinery without raising the boom. Repeat this test using the secondary drum clutch. An inspection shall be made for ease of control operation, smoothness of load lowering, and machinery operation. This operation shall be performed without binding, chattering of machinery, slipping at the clutches, or unusual effort of the operator (see 3.15.7.1).

4.4.11 Sound level testing.

4.4.11.1 Interior sound levels. The noise level at the operator's station shall be tested in accordance with SAE J919. The data shall include dBA and octave band pressure levels taken at the operator's station. Duty cycle testing shall not be used.

4.4.11.2 Exterior sound levels. The crane shall be tested in accordance with SAE J88.

4.4.12 Piledriving. When applicable (see 6.2), attach the Government-loaned pile-driving leads, catwalk, and hammer to the boom and revolving superstructure. Perform pile-driving operations for 4 hours. Inability to operate with pile-driving hammer, malfunction, evidence of any damage, or failure of any component shall constitute failure of this test.

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4.4.13 Backhoe and shovel-front.

4.4.13.1 Backhoe. When furnished with a backhoe attachment (see 6.2), perform maximum capacity heavy-duty trenching operations in undisturbed soil including digging depth specified in table III for not less than 4 hours. Failure to perform as specified, any evidence of malfunction, damage, or permanent deformation shall constitute failure of this test.

4.4.13.2 Shovel-front attachment. When furnished with a shovel-front attachment (see 6.2) perform shovel-front excavating operations at maximum capacity, and include equal portions of time operating at each of the ranges specified in table II. The total period of operation shall be not less than 4 hours. The operations shall be conducted in a gravel pit or rock quarry. Failure to perform as specified, any evidence of malfunction, damage, or permanent deformation shall constitute failure of this test.

4.4.14 Wrecking. Perform simulated wrecking operations both in the horizontal and vertical planes for a total of 4 hours. Failure or inability to operate with wrecking ball, malfunction, evidence of any damage, or failure of any component shall constitute failure of this test.

4.4.15 Clamshell and dragline. Perform maximum-capacity, heavy-duty clamshell and dragline operations in undisturbed heavy clay soil. The operation shall be performed with the clamshell and dragline bucket for at least 8 hours (4 hours for each bucket). Inability to operate with clamshell and dragline buckets, inability to complete the test, malfunction, evidence of any damage, or failure of any component shall constitute failure of this test.

4.4.16 Measurement of electromagnetic radiation. The electromagnetic radiation shall be measured in accordance with SAE J551.

4.4.17 Lifting and tie down attachment test or structural analysis. To determine that the lifting and tiedown attachments conform to 3.8.1, the contractor shall either test the attachments or furnish a complete structural analysis of the lifting and tiedown provisions. Weld failure, damage, or permanent deformation during test or an incomplete analysis or a determination that the provisions are inadequate shall be cause for rejection of the first article.

4.4.18 Test report. The manufacturer's test report must be provided (see 6.2) to document the results of the first article test and inspection and to determine compliance with the requirements of this specification. In addition to the test results, the manufacturer must provide the following certifications.

(a) Conformance to OSHA Regulations, EPA Standards, and PCSA Std No. 1 (see 3.1 and 3.6.1).

(b) Crane hook - The crane manufacturer shall furnish certification from the crane hook manufacturer, that the hook furnished with the

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crane is of compatible size, having the proper load capacity and safety factor.

(c) Stability - Certification that the crane stability was tested in accordance with SAE J765.

(d) Ground bearing pressure - Calculations or other certification indicating the ground bearing pressure for the crane.

(e) Electromagnetic interference - Certification that the electromagnetic radiation from the crane is within the limits of SAE J551.

(f) Lifting and tiedown attachments - The manufacturer shall provide an engineering analysis certifying the adequacy of the lifting and tiedown attachments provided.

(g) Stress - Certification that the stress levels of the crane do not exceed the recommendation of SAE J987 and that the test results are available upon request.

(h) Engine - Certification that the engine furnished has been approved for the application by the engine manufacturer.

(i) Clutch and torque converter - Certification that the clutch and torque converter furnished have been approved for the application by the clutch and torque converter manufacturers.

4.5 Production tests. Operational acceptance tests shall be made on production units and shall be in accordance with ANSI B30.5 procedure conducted prior to shipping.

4.6 Inspection of packaging. The preservation, packing and marking shall be examined to determine compliance with the Quality Assurance Provisions of MIL-C-3580 and the specific requirements specified in Section 5 of this document.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or C as specified (see 6.2).

5.1.1 Level A. Each complete crane, components, and attachments shall be preserved in accordance with the level A preservation requirements of MIL-C-3580 except engine cooling systems shall be preserved by the water and antifreeze procedure specified in MIL-E-10062, using antifreeze conforming to MIL-A-46153.

5.1.2 Level B. Each complete crane, components, and attachments shall be preserved in accordance with the level B preservation requirements of MIL-C-3580 except engine cooling systems shall be preserved by the water and antifreeze conforming to MIL-A-46153.

5.1.3 Level C. Each complete crane and all items issued with the crane shall be preserved for retail distribution directly to a using customer. In addition, the following specific requirements shall be accomplished. Items with critical surfaces required for fit or function shall be provided protection by the use of preservative coating, volatile corrosion inhibitors, or desiccated packs. Items requiring protection from physical damage to surfaces, or which are fragile by nature, shall be protected by wrapping, pack compartmentization, or cartonizing

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of the individual item. The cooling system shall be filled with a fresh solution of 50 percent water and 50 percent antifreeze, conforming to MIL-A-46153, and a tag attached indicating the temperature to which the cooling system can be subjected to before freezing. The crankcase shall contain the required amount of lubricating oil. A guide shall be furnished indicating the preservation and packing materials to be removed, in detail, and any other servicing required prior to placing the crane in operation.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Each complete crane preserved as specified in 5.1, shall be packed in accordance with the level A packing requirements of MIL-C-3580.

5.2.2 Level B. Each complete crane, preserved as specified in 5.1, shall be packed in accordance with the level B packing requirements of MIL-C-3580.

5.2.3 Level C. Each complete crane, preserved as specified in 5.1, shall be prepared for mobile shipment in a manner to assure carrier acceptance and safe delivery to destination at the lowest ratings of the carrier. Crane components and the attachments removed to reduce cubage or removed to prevent damage shall be secured to the crane prior to shipment. Removed crane components and attachments that cannot be secured to the crane shall be properly marked to insure shipment at the same time as the crane.

5.3 Marking. Marking for shipment and storage shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The crane, together with the various items of front-end operating equipment, are intended for use in handling materials, lifting and transporting loads short distances, soil stripping, earthmoving, ditch digging, pile driving, and demolition.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) When powered load lowering is required on the primary and secondary load hoist drums (see 3.15.5).
(c) When an overload indicating system is required (see 3.17.1.7).
(d) When boom extensions are required, lengths and number required (see 3.17.2.1).
(e) When boom jib is required and length required (see 3.17.2.2).
(f) When clamshell bucket is required (see 3.17.2.3).
(g) When tagline is required (see 3.17.2.4).
(h) When dragline bucket is required (see 3.17.2.5).
(i) When dragline fairleads are required (see 3.17.2.6).
(j) When a wrecking ball is required and size required (see 3.17.2.7).
(k) When the pile driver leads and catwalk are required (see 3.17.2.8).
(l) When the piledriver hammer is required and size required (see 3.17.2.9).

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(m) When the shovel front attachment is required (see 3.17.2.10).
(n) When the backhoe attachment is required (see 3.17.2.11).
(o) When an auxilary or third drum is required (see 3.15.7.3).
(p) When the Government loaned property is not required (see 3.24).
(q) Number of first article units and time frame required for submission (see 4.3).
(r) Time frame for submitting first article test report (see 4.4.18).
(s) Level of preservation and packaging and level of packing required (see 5.1 and 5.2).

6.3 Government-furnished property. The contracting officer should arrange to furnish the contractor the property listed in 3.23.

6.4 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.24 unless the item is being purchased under the same contract.

6.5 Technical information package. The TIP may be obtained from US Army Mobility Equipment Research and Development Command, ATTN: DRDME-HK, Fort Belvoir, VA 22060.

6.6 Vehicle classification. The vehicle classification data referred to in 6.7 should be entered on the Vehicle Classification Data Sheet which may be obtained from US Army Mobility Equipment Research and Development Command, ATTN: DRDME-HK, Fort Belvoir, VA 22060.

6.7 Data requirements. The contracting officer should include requirements for such technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manuals and provisioning requirements for repair parts and maintenance tools to be furnished.

6.8 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.4).

6.9 The contracting officer should arrange to furnish the warning decal set, FSN 7690-267-5778, and provide mounting instructions as required in 3.19.1.

Custodian:
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Preparing activity:
Army - ME

User activity:
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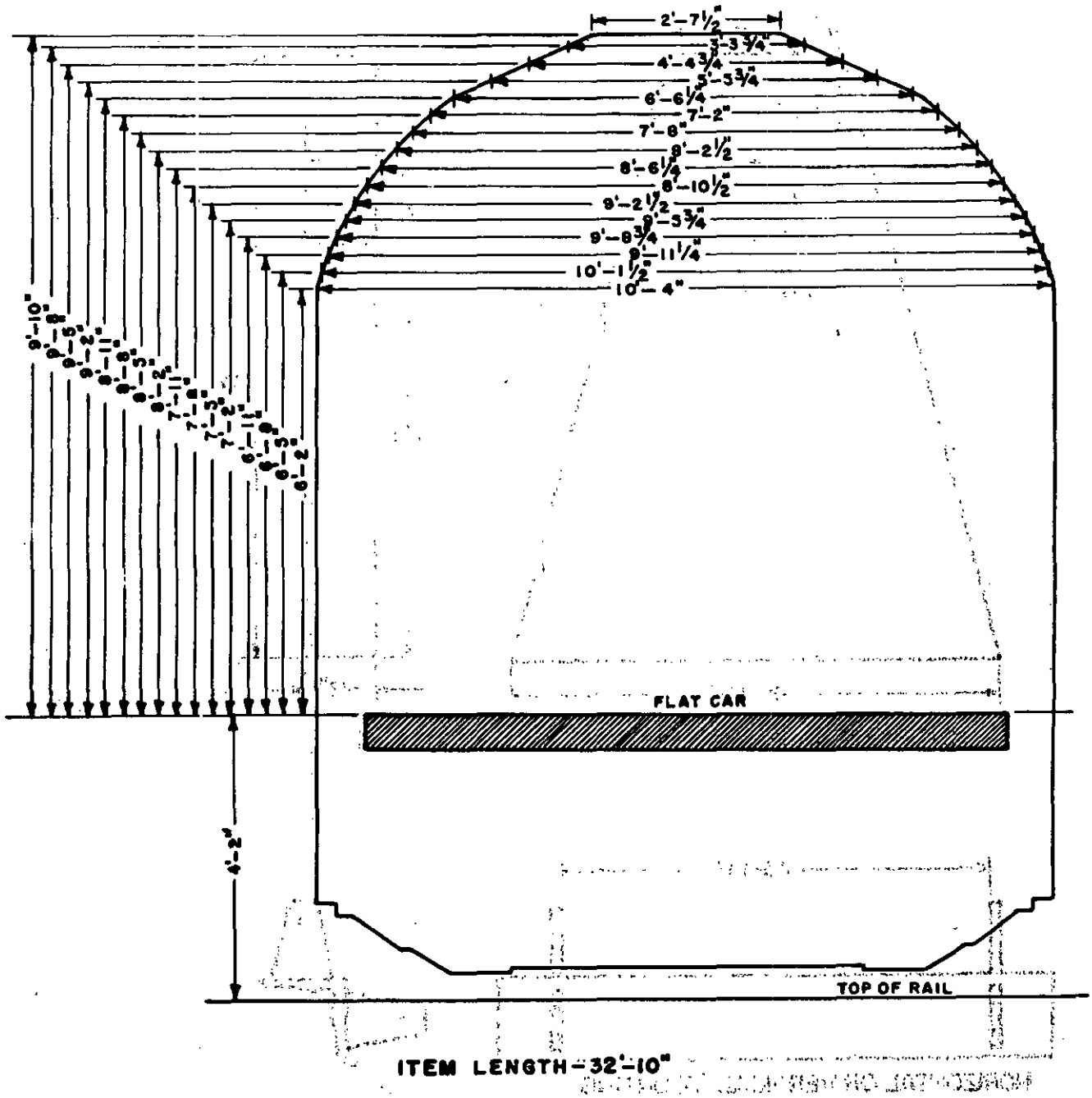
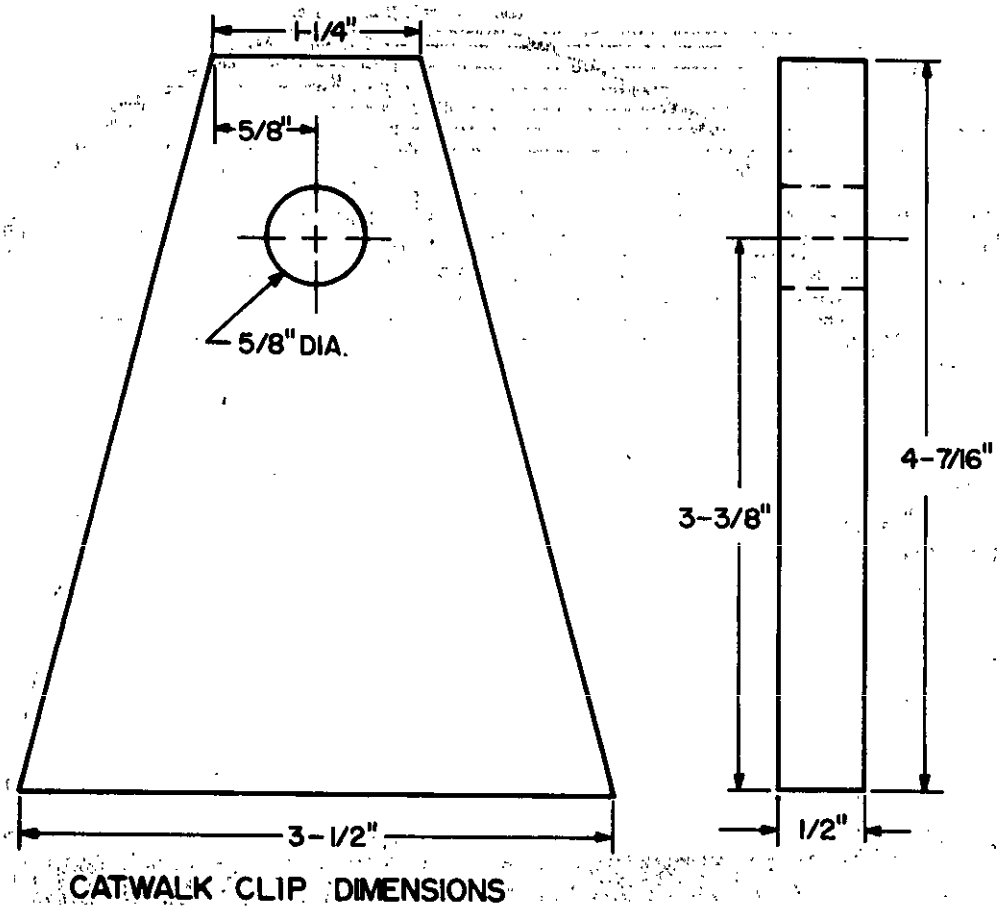


Figure 1. BERNE INTERNATIONAL (CLEARANCE DIAGRAM)

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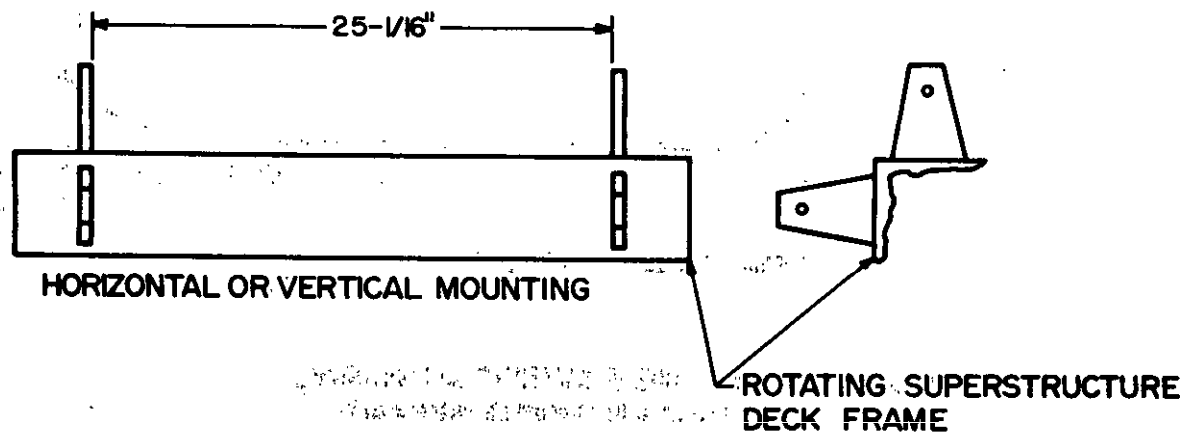
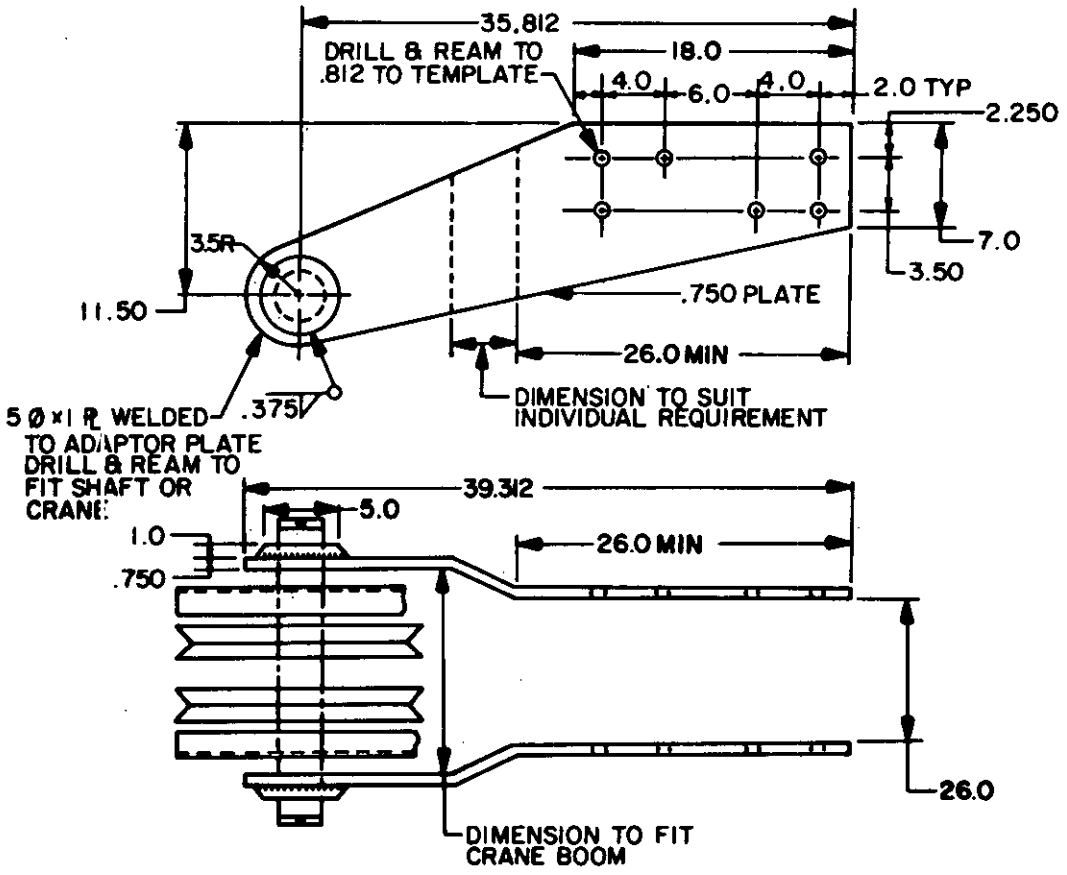


FIGURE 2. CATWALK CONNECTION CLIP

X-2959

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NOTE: ALL DIMENSIONS ARE IN INCHES.

FIGURE 3. Piledriver lead adapter.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-C-52648C(ME) Crane, Crawler Mounted, 40 Ton at 12 Foot Radius, 2 Cubic Yard, Diesel-Engine-Driven

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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